

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Research: Agentic computer-use with execution verification – Multimodal model routing – Production agentic RAG

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (Machine Learning), GPA: 4.0/4.0

Aug 2024 – Dec 2026

– **Coursework:** Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, CGPA: 8.8/10.0

Aug 2016 – Aug 2020

RESEARCH EXPERIENCE

Georgia Tech – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Building a closed-loop agentic pipeline for parametric CAD code generation: LLM generates CadQuery, execution feedback drives LangGraph repair loop with geometric verification (Chamfer/Hausdorff on STL/B-Rep) – inference-time search with verifiable reward, 16 modules, 202 tests, VLM visual judgment.

Georgia Tech – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **ARTEMIS:** Cost-aware VLM routing with trained neural router, SLA-aware load balancing, 6+ backends (Gemma-3, Qwen-VL, DeepSeek-OCR, Llama-4) via vLLM; dynamic accuracy/cost/latency modes.
- **SHASTRA:** Trace-to-graph pipeline converting GAIA agent traces (94 sessions, 4K events) into reusable task-composition graphs; LangGraph planner with Pydantic registry for workflow reuse under cost/latency constraints.
- **ATHENA:** 5-agent supervisor-worker orchestration via LangGraph Command; reflection-based critique, dynamic plan modification, Pydantic structured outputs; cross-modal eval (BLEU-4 +0.18, CLIPScore) on 100 videos.

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Led agentic RAG diagnostics: LLM plans multi-step tool calls over telemetry, verifies hypotheses via structured function calling, produces root-cause analysis – reduced resolution time 70% (hackathon to production).
- Scaled OpenSearch ingestion 50→2,000 events/sec (40x) with Golang; deployed edge ML via ONNX Runtime (40% latency reduction); built SLA forecasting for 60+ classifiers.
- Patent PCT/IN2022/058026: Unsupervised distributional thresholding for wireless anomaly detection; cut manual troubleshooting 75%.

PUBLICATIONS & PATENTS

Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless anomaly detection.

TECHNICAL SKILLS

Agentic AI: LangChain, LangGraph, Agentic RAG, Tool/Function Calling, Multi-Agent Orchestration, Supervisor-Worker, Planning & Reasoning, LLM-as-a-Judge, Structured Generation (Pydantic)

LLMs / VLMs: vLLM, VLM Routing & Evaluation, CLIP, SBERT, FAISS, Inference Optimization, Model Evaluation, Cross-Modal Retrieval, Multimodal (video, audio, image, text)

AI / ML: PyTorch, Hugging Face Transformers, Deep Learning, CNNs, Vision Transformers, NLP, Computer Vision, Scikit-Learn, NumPy, Pandas

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)

Systems: Python, Go, C/C++, SQL/PostgreSQL, Redis, OpenSearch/Elasticsearch, Docker, Linux, Git, ONNX Runtime, FastAPI, Azure, W&B, TensorBoard